Part 5 - Portfolio Analytics

Summary. MSCI USA Value delivers strong HML exposure (+0.322; t = 19.7), but no statistically significant alpha (t = -0.91). A fitted Fama-French replication shows a higher historical Sharpe (0.529 versus 0.450), subject to trading costs and in-sample bias. The multi-factor index has a weak benchmark-relative information ratio (0.045). Minimum Volatility has the strongest stand-alone Sharpe (0.66) but meaningful benchmark-relative risk.

5A. MSCI Value and Fama-French value definitions

MSCI uses three value indicators: book value/price, 12-month forward earnings/price, and dividend yield. It separately measures growth using long- and short-term forward EPS growth, current internal growth, historical long-term EPS growth, and historical long-term sales-per-share growth. Value and growth membership can overlap, with partial style weights. MSCI's exceptions apply **only to growth variables** and are set by GICS industry and by size segment: long-term historical sales-per-share growth is not used for Banks and Diversified Financials (except Multi-Sector Holdings and Financial Exchanges & Data), and long-term forward EPS growth is not used for small-cap indexes. In those cases the remaining growth variables are averaged.

Fama-French HML instead uses **one cheapness measure, book-to-market (B/M)**. Stocks are sorted into a 2-by-3 grid by size (small/big) and B/M (low/neutral/high), rebalanced each June. HML is the average return of high-B/M portfolios minus low-B/M portfolios across the two size groups. The low-B/M end represents growth without MSCI's separate five-signal growth score.

Design choice	MSCI USA Value	Fama-French HML
Value signal	Book/price, 12-month forward earnings/price, dividend yield	Single book-to-market ratio
Growth treatment	Separate five-indicator score	Low book-to-market end of sort
Construction	Long-only large/mid-cap style weights	Long-short 2x3 size/B/M sorts; June rebalance
Implication	Investable but blends style and universe decisions	Pure value spread; long/short, not directly investable

Exhibit 5A. Value methodology comparison, US equity universe (MSCI USA Standard; Fama-French US factor portfolios). Sources: MSCI Global Investable Market Value and Growth Index Methodology (February 2021); Kenneth R. French Data Library, factor definitions.

5B. Factor exposure regression

Using 343 monthly gross USD returns (January 1995-July 2023), we regress MSCI USA Value on an intercept, RF, market excess return, HML and SMB. Coefficients measure exposure; conventional OLS t-statistics measure estimation precision.

Regressor	Loading	t-statistic	Interpretation
Intercept (monthly %)	-0.0713	-0.91	No significant positive alpha
RF	0.8646	2.79	Cash-rate exposure estimated, not fixed
Market excess	0.9480	76.99	Near-market equity sensitivity
HML	+0.3216	19.68	Strong value exposure
SMB	-0.1744	-9.90	Large-cap bias embedded in style index

Exhibit 5B. OLS factor fingerprint of MSCI USA Value, January 1995-July 2023, 343 monthly observations, MSCI USA Standard and Fama-French US factors. Source: supplied MSCI and Fama-French workbook.

R-squared is **94.84%**. In a month when HML gains one percentage point, the fitted index return increases by 0.322 percentage points, holding other inputs fixed; a one-point SMB gain reduces it by 0.174 points. Thus MSCI Value is not pure HML: its large-cap tilt creates a distinct performance headwind when small caps outperform. **Verdict:** MSCI Value behaves as advertised: strong value exposure (HML +0.32; t = 19.7) alongside a large-cap tilt (SMB -0.17; t = -9.9). The intercept does not establish additional alpha.

5C. Does the methodology enhance results

The monthly intercept is **-0.0713%** (t = -0.91; p = 0.366), approximately -0.86% a year by simple annualization. There is therefore no statistically significant alpha beyond the fitted factors; the point estimate is negative. After the **May 2003 methodology change**, the June 2003-July 2023 subsample gives an intercept of **-0.0378% a month (t = -0.55; n = 242)**, which supports the same conclusion.

Sample	Intercept (monthly %)	t-statistic	Months
Jan 1995-Jul 2023	-0.0713	-0.91	343
Jun 2003-Jul 2023	-0.0378	-0.55	242

Exhibit 5C. Intercept of MSCI USA Value on RF, Mkt-RF, HML and SMB, full sample and after the May 2003 methodology change, gross USD. Source: supplied workbook; authors' OLS calculations.

5D. Fund replication

The fitted zero-alpha replica is 0.8646 RF + 0.9480 (Mkt-RF) + 0.3216 HML - 0.1744 SMB. Replication was effective in sample (correlation **0.974**), but the apparent performance advantage is not a tradeable, out-of-sample result.

Metric	MSCI Value	FF replication
Arithmetic return, annualized	9.87%	10.73%
CAGR	9.05%	10.05%
Volatility, annualized	15.27%	14.87%
Geometric Sharpe	0.450	0.529

Exhibit 5D.1. Return, risk and Sharpe ratio of MSCI USA Value and the fitted Fama-French replication, January 1995-July 2023, gross USD, in sample. Sharpe = (CAGR - compounded RF) / annualized volatility. Source: supplied workbook; authors' calculations.

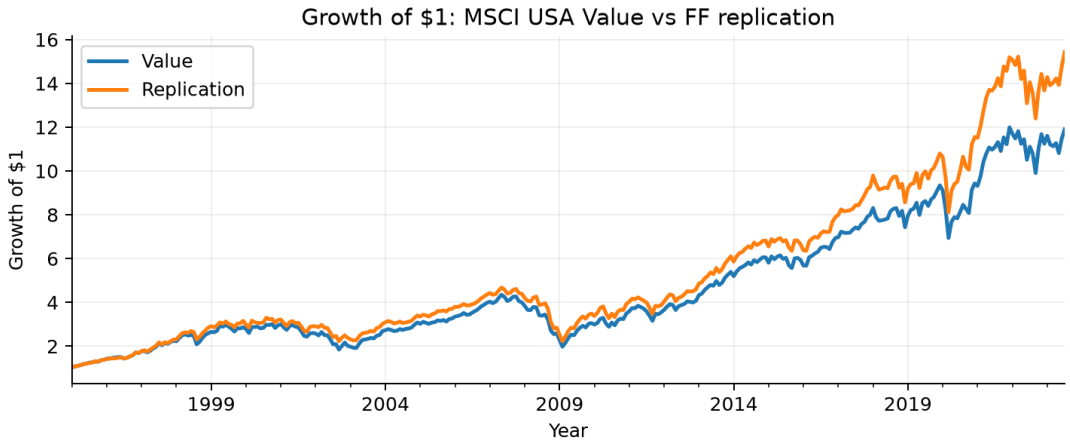


Exhibit 5D.2. Growth of $1, MSCI USA Value versus fitted Fama-French replication, January 1995-July 2023, gross USD, in sample. Source: supplied workbook; authors' calculations.

Before implementation, check shorting availability and trading/financing costs for long-short factors, replication fees versus a low-cost index fund, and the risk of in-sample coefficient fitting. The 0.86 percentage-point arithmetic return gap mirrors the insignificant -0.86% a year fitted intercept.

5E. Benchmark relative performance

Active return is each index's monthly return less MSCI USA Standard; the multi-factor index equally weights four styles and rebalances monthly. Tracking error is the annualized standard deviation of monthly active returns (monthly standard deviation x sqrt(12)); the information ratio (IR) is annualized arithmetic active return divided by tracking error.

Index	Active return (annual)	Tracking error	IR
Value	-1.34%	5.15%	-0.261
Minimum Volatility	-0.80%	6.38%	-0.125
Equal Weight	+0.59%	5.40%	0.109
Momentum	+2.16%	7.73%	0.279
Multi-factor index	+0.15%	3.39%	**0.045**

Exhibit 5E.1. Annualized active return, tracking error and information ratio versus MSCI USA Standard, January 1995-July 2023, gross USD; MSCI USA Value, Minimum Volatility, Equal Weighted, Momentum and the multi-factor index. Source: supplied MSCI index data; authors' calculations.

The multi-factor index's IR of 0.045 indicates little active return per unit of benchmark-relative risk. Its 0.15% arithmetic active return and 0.34 percentage-point geometric CAGR advantage measure different quantities. As Part 1 found, the index nevertheless beat the market on Sharpe (0.60 versus 0.54), primarily because of lower volatility.

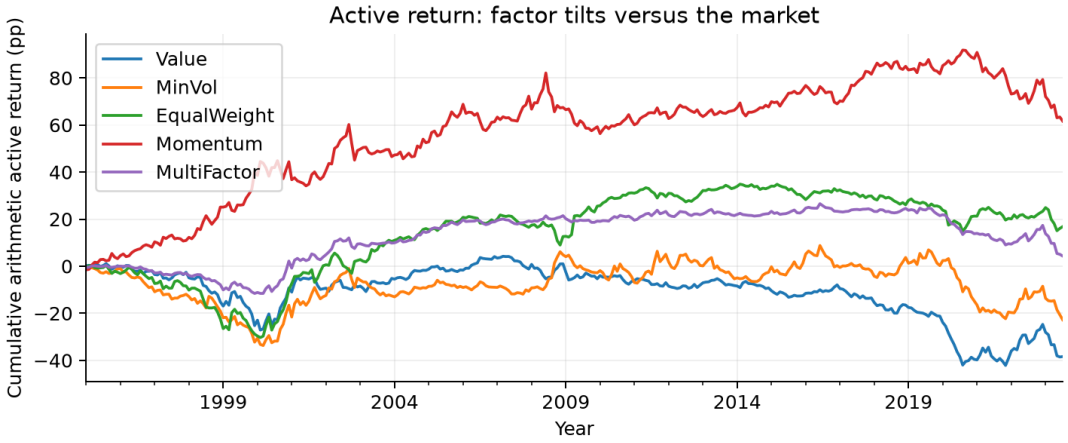


Exhibit 5E.2. Cumulative arithmetic active return versus MSCI USA Standard, January 1995-July 2023, gross USD; "MultiFactor" in the legend is the multi-factor index. Source: supplied MSCI index data; authors' calculations.

5F. Active risk decomposition and investment decision

Exhibit 5F splits the multi-factor index's active-return variance across the four indexes. With equal-weight active returns a_i and index active return a_p = (1/4) sum_i a_i, each index's share is Cov(a_i, a_p) / [4 Var(a_p)], and the shares sum to 100%.

Factor	Share of active-return variance
Value	23.26%
Minimum Volatility	34.08%
Equal Weight	17.86%
Momentum	24.81%

Exhibit 5F. Covariance contributions to multi-factor index active-return variance, January 1995-July 2023, MSCI USA smart-beta indexes relative to MSCI USA Standard, gross USD. Source: supplied MSCI index data; authors' calculations.

Minimum Volatility is the largest active-risk driver at 34.08%, despite equal portfolio weights.

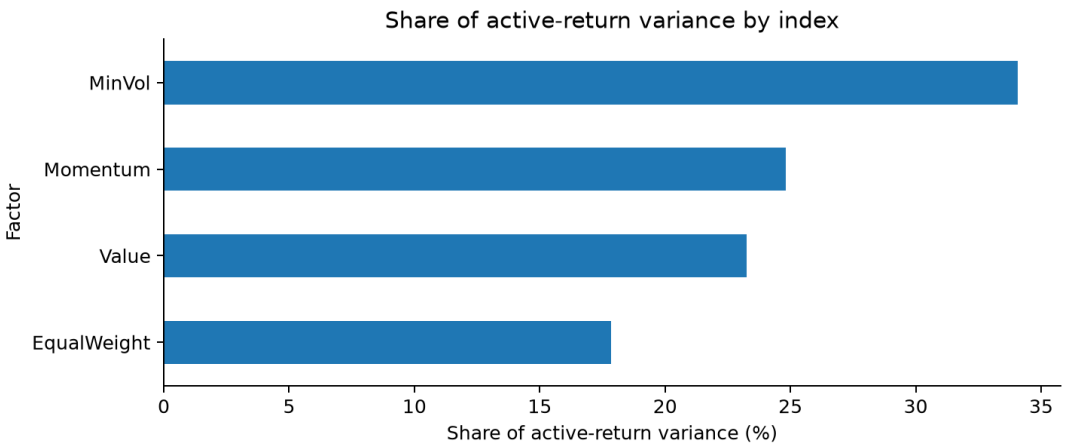


Exhibit 5F.2. Share of multi-factor index active-return variance by index, January 1995-July 2023, gross USD. Source: supplied MSCI index data; authors' calculations.

Decision: Minimum Volatility is the preferred stand-alone risk-adjusted investment (Sharpe 0.66), but its IR of -0.125, tracking error of 6.38%, and market beta near 0.72 make it harder to hold for benchmark-constrained managers during strong equity rallies. The multi-factor index is a defensible lower-volatility compromise, not a compelling source of active alpha.

Methods and caveats. Monthly gross USD observations cover January 1995-July 2023. OLS t-statistics are conventional, without autocorrelation adjustment. Factor replication is fitted and assessed in sample; implementation costs, shorting constraints, and fees are excluded.